

Payments Platform – Solution Architecture Overview

Document Control

Document ID: PAY-ARCH-014 | Version: 1.8 | Owner: Payments Engineering | Status: Approved

Overview

The payments platform provides API-led transaction processing for payment initiation, validation, fraud screening, core-banking posting and downstream notifications.

Components

API Gateway, Payment Orchestrator, Account Validation Service, Fraud Screening Service, Core Banking Adapter, Transaction Store, Event Bus, Notification Service and Audit/Monitoring Services.

Transaction Flow

Client → API Gateway → Payment Orchestrator → Account Validation → Fraud Screening → Core Banking Adapter → Transaction Store → Event Bus → Notification.

Integration

REST/JSON is used for synchronous operations. Events are used for notifications and analytics. Correlation IDs are propagated across service boundaries.

NFRs

Target availability is 99.9%. Standard requests should normally complete within two seconds under agreed operating conditions. State transitions require audit records.

Dependencies

Core Banking Platform, Fraud Screening Platform, Notification Platform, Enterprise Identity Service, Message Bus and Monitoring Platform.

Risks

Legacy integration latency, duplicate event processing, downstream timeouts, peak-volume throttling and transaction-state reconciliation.